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Grade 13.00 out of 15.00 (86.67%)

Question 1 | Correct Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- ☐ a. In the entire list
- ☐ b. In the right sublist
- ☐ c. At the beginning of the list
- ☒ d. In the left sublist ✓

Question 2 | Correct Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- ☐ a. $O(\log n)$
- ☐ b. $O(n)$
- ☐ c. $O(n \log n)$
- ☒ d. $O(1)$ ✓

Question 3 | Incorrect Mark 0.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- ☒ a. Number values should only be present ✗
Number values should only be present
- ☐ b. The list must be sorted
- ☐ c. There must be a mechanism to delete and/or insert elements in the list
- ☐ d. There should be direct access to the middle element in any sublist

Question 4 | Incorrect Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- ☒ a. The lower middle element is chosen as the middle element ✖
- ☐ b. The middle element is chosen randomly
- ☐ c. The higher middle element is chosen as the middle element
- ☐ d. The search stops

Question 5 | Correct Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- ☐ a. Hashing
- ☐ b. Dynamic search
- ☐ c. Binary search
- ☒ d. Linear search ✔

Question 6 | Correct Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- ☐ a. The search continues in the right sublist
- ☒ b. The search ends successfully ✔
- ☐ c. The search continues in the left sublist
- ☐ d. The list is sorted

Question 7 | Correct Mark 1.00 out of 1.00

Given an array `arr = {45,77,89,90,94,99,100}` and `key = 100`; What are the mid values(corresponding array elements) generated in the first and second iterations?

- ☐ a. 90 and 100
- ☒ b. 90 and 99 ✔
- ☐ c. 89 and 94
- ☐ d. 94 and 99

Question 8 | Correct Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- ☐ a. Item is the last element in the array or item is not there at all
- ☒ b. When the item is somewhere in the middle of the array ✓
- ☐ c. When the item is the last element in the array
- ☐ d. When the item is not the array at all

Question 9 | Correct Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- ☐ a. 89 and 94
- ☒ b. 90 and 99 ✓
- ☐ c. 89 and 99
- ☐ d. 90 and 94

Question 10 | Correct Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- ☐ a. 1000
- ☐ b. 100
- ☐ c. 500
- ☒ d. 10 ✓

Question 11 | Correct Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- ☐ a. Discovering
- ☐ b. Finding
- ☒ c. Searching ✓
- ☐ d. Mining

Question 12 | Correct Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n ?

- ☐ a. $\log n$
- ☐ b. 1
- ☐ c. $n/2$
- ☒ d. n ✓

Question 13 | Correct Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- ☐ a. $O(1)$
- ☒ b. $O(n)$ ✓
- ☐ c. $O(n \log n)$
- ☐ d. $O(\log n)$

Question 14 | Correct Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- ☐ a. Interpolation search
- ☐ b. Binary search
- ☒ c. Linear search ✓
- ☐ d. Hash search

Question 15 | Correct Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- ☐ a. In the left sublist
- ☒ b. In the right sublist ✓
- ☐ c. In the middle of the list
- ☐ d. At the beginning of the list